do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Configure Authentication and Authorization

Configure the HTPasswd identity provider, create groups, and assign roles to users and groups.

Outcomes

- Create users and passwords for HTPasswd authentication.
- Configure the identity provider for HTPasswd authentication.
- Assign cluster administration rights to users.
- Create groups and add users to groups.
- Manage user privileges in projects by granting privileges to groups.

As the `student` user on the `workstation` machine, use the `lab` command to prepare yo environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start auth-review
```

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Instructions

The API URL of your OpenShift cluster is `https://api.lab.example.com:6443`, and the `oc` command is already installed on your `workstation` machine.

1. Update the existing `~/DO280/labs/auth-review/tmp_users` HTPasswd authentication file to remove the `analyst` user. Ensure that the `tester` and `leader` users in the file use `L@bR3v!ew` as the password. Add two entries to the file for the `new_admin` and `new_developer` users. Use `L@bR3v!ew` as the password for each new user.

   1.1. Remove the `analyst` user from the `~/DO280/labs/auth-review/tmp_users` HTPasswd authentication file.

   ```
   [student@workstation ~]$ htpasswd -D ~/DO280/labs/auth-review/tmp_users analyst
   Deleting password for user analyst
   ```

   1.2. Update the entries for the `tester` and `leader` users to use `L@bR3v!ew` as the password. Add entries for the `new_admin` and `new_developer` users with `L@bR3v!ew` as the password.

   ```
   [student@workstation ~]$ for NAME in tester leader new_admin new_developer ; \
   do \
   htpasswd -B -b ~/DO280/labs/auth-review/tmp_users ${NAME} 'L@bR3v!ew' ; \
   done
   Updating password for user tester
   Updating password for user leader
   Adding password for user new_admin
   Adding password for user new_developer
   ```

   1.3. Review the contents of the `~/DO280/labs/auth-review/tmp_users` file. This file does not contain a line for the `analyst` user. The file includes two new entries with hashed passwords for the `new_admin` and `new_developer` users.

   ```
   tester:$2y$05$S6KfxhuGRTBXOkhjuF00j.163yqDcQbJKbpnV1HlNKtTexuiRdi9.
   leader:$2y$05$Ufg.vA6wacDyWaPiOJ1LI.cgYJrOj0G48sIZIyTqMqYA/jV1/qx0i
   new_admin:$2y$05$egylVnEchsSenCI.iqA06OifodJesZt8JHDLAjFrouyVHls9otu0q
   new_developer:$2y$05$dDa/Tai9aDB5b7FuQZgw9.XUmYAT.9IjoqWzinK9Ode9oFfVK
   IcC.
   ```

   Hide Solution

2. Log in to your OpenShift cluster as the `admin` user with `redhatocp` as the password. Configure your cluster to use the HTPasswd identity provider by using the defined user names and passwords in the `~/DO280/labs/auth-review/tmp_users` file. For grading,

use the `auth-review` name for the secret.

2.1. Log in to the cluster as the `admin` user.

```
[student@workstation ~]$ oc login -u admin -p redhatocp \
https://api.lab.example.com:6443
Login successful.

...output omitted...
```

2.2. Create an `auth-review` secret by using the `~/DO280/labs/auth-review/tmp_users` file.

```
[student@workstation ~]$ oc create secret generic auth-review \
--from-file htpasswd=/home/student/DO280/labs/auth-review/tmp_users \
-n openshift-config
secret/auth-review created
```

2.3. Export the existing OAuth resource to the `~/DO280/labs/auth-review/oauth.yaml` file.

```
[student@workstation ~]$ oc get oauth cluster \
-o yaml > ~/DO280/labs/auth-review/oauth.yaml
```

2.4. Edit the `~/DO280/labs/auth-review/oauth.yaml` file to add the `htpasswd` identity provider in addition to the existing `ldap` identity provider. Ensure that the `htpasswd`, `mappingMethod`, `name`, and `type` fields are at the same indentation level.

```
apiVersion: config.openshift.io/v1
kind: OAuth
...output omitted...
spec:
  identityProviders:
  - ldap:
      attributes:
...output omitted...
    name: Red Hat Identity Management
    type: LDAP
  - htpasswd:
      fileData:
        name: auth-review
    mappingMethod: claim
    name: htpasswd
    type: HTPasswd
```

NOTE

For convenience, the `~/DO280/solutions/auth-review/oauth.yaml` file contains a minimal version of the OAuth configuration with the specified customizations.

2.5. Apply the customized resource that you defined in the previous step.

```
[student@workstation ~]$ oc replace -f ~/DO280/labs/auth-review/oauth.yaml
oauth.config.openshift.io/cluster replaced
```

2.6. A successful update to the `oauth/cluster` resource re-creates the `oauth-openshift` pods in the `openshift-authentication` namespace. Wait until the new `oauth-openshift` pods are ready and running, and the previous pods have terminated.

```
[student@workstation ~]$ watch oc get pods -n openshift-authentication
Every 2.0s: oc get pods -n openshift-authentication         workstation: Fri Jul 31 08:19:42 2026

NAME                               READY   STATUS    RESTARTS   AGE
oauth-openshift-67576dc855-kfs6p   1/1     Running   0          29s
```

Press **Ctrl**+**C** to exit the `watch` command.

NOTE

Pods in the `openshift-authentication` namespace redeploy when the `oc replace` command succeeds.

In this exercise, changes to authentication might require a few minutes to apply.

You can examine the status of pods and deployments in the `openshift-authentication` namespace to monitor the authentication status. You can also examine the `authentication` cluster operator for further status information.

Provided that the previously created secret is created correctly, you can log in by using the HTPasswd identity provider.

Hide Solution

3. Make the `new_admin` user a cluster administrator. Log in as both the `new_admin` and `new_developer` users to verify HTPasswd user configuration and cluster privileges.

3.1. Assign the `new_admin` user the `cluster-admin` role.

```
[student@workstation ~]$ oc adm policy add-cluster-role-to-user \
cluster-admin new_admin
Warning: User 'new_admin' not found
clusterrole.rbac.authorization.k8s.io/cluster-admin added: "new_admi
n"
```

> **NOTE**
>
> You can safely ignore the warning that the `new_admin` user is not found.

3.2. Log in to the cluster as the `new_admin` user to verify that HTPasswd authentication is configured correctly.

```
[student@workstation ~]$ oc login -u new_admin -p 'L@bR3v!ew'
Login successful.

...output omitted...
```

3.3. Use the `oc get nodes` command to verify that the `new_admin` user has the `cluster-admin` role. The names of the nodes from your cluster might be different.

```
[student@workstation ~]$ oc get nodes
NAME       STATUS   ROLES                         AGE   VERSION
master01   Ready    control-plane,master,worker   14d   v1.35.5
```

3.4. Log in to the cluster as the `new_developer` user to verify that the HTPasswd authentication is configured correctly.

```
[student@workstation ~]$ oc login -u new_developer -p 'L@bR3v!ew'
Login successful.

...output omitted...
```

3.5. Use the `oc get nodes` command to verify that the `new_developer` user does not have cluster administration privileges.

```
[student@workstation ~]$ oc get nodes
Error from server (Forbidden): nodes is forbidden: User "new_develope
r" cannot list resource "nodes" in API group "" at the cluster scope
```

Hide Solution

4. As the `new_admin` user, create a `managers` group, and add the `leader` user to the group. As the `leader` user, create the `auth-review` project.

4.1. Log in to the cluster as the `new_admin` user.

```
[student@workstation ~]$ oc login -u new_admin -p 'L@bR3v!ew'
Login successful.

...output omitted...
```

4.2. Create a `managers` group.

```
[student@workstation ~]$ oc adm groups new managers
group.user.openshift.io/managers created
```

4.3. Add the `leader` user to the `managers` group.

```
[student@workstation ~]$ oc adm groups add-users managers leader
group.user.openshift.io/managers added: "leader"
```

4.4. As the `leader` user, create the `auth-review` project.

```
[student@workstation ~]$ oc login -u leader -p 'L@bR3v!ew'
Login successful.

...output omitted...
```

The user who creates a project is automatically assigned the `admin` role on the project.

```
[student@workstation ~]$ oc new-project auth-review
Now using project "auth-review" on server "https://api.lab.example.co
m:6443".

...output omitted...
```

Hide Solution

5. Create a `developers` group and grant edit privileges on the `auth-review` project. Add the `new_developer` user to the group.

5.1. Log in to the cluster as the `new_admin` user.

```
[student@workstation ~]$ oc login -u new_admin -p 'L@bR3v!ew'
Login successful.

...output omitted...
```

5.2. Create a `developers` group.

```
[student@workstation ~]$ oc adm groups new developers
group.user.openshift.io/developers created
```

5.3. Add the `new_developer` user to the `developers` group.

```
[student@workstation ~]$ oc adm groups add-users developers new_devel
oper
group.user.openshift.io/developers added: "new_developer"
```

5.4. Grant edit privileges to the `developers` group on the `auth-review` project.

```
[student@workstation ~]$ oc policy add-role-to-group edit developers
clusterrole.rbac.authorization.k8s.io/edit added: "developers"
```

Hide Solution

6. Create a `qa` group and grant view privileges on the `auth-review` project. Add the `tester` user to the group.

6.1. Create a `qa` group.

```
[student@workstation ~]$ oc adm groups new qa
group.user.openshift.io/qa created
```

6.2. Add the `tester` user to the `qa` group.

```
[student@workstation ~]$ oc adm groups add-users qa tester
group.user.openshift.io/qa added: "tester"
```

6.3. Grant view privileges to the `qa` group on the `auth-review` project.

```
[student@workstation ~]$ oc policy add-role-to-group view qa
clusterrole.rbac.authorization.k8s.io/view added: "qa"
```

Hide Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade auth-review
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish auth-review
```

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